



COSC 4337 | Data Science II

Section: 15555 | Face-to-Face

Prerequisite(s): A grade of C- or better in COSC 3337 and declared COSC major, COSC minor, or Data Science minor. Emphasis on practical techniques for working with large-scale data. Advanced machine learning topics such as Neural Networks and Deep Learning, ensemble learning approaches and model evaluation.

Instructor Information

Dr. Nouhad Rizk

Department: Computer Science Department

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Office Hours

9:00 AM - 10:00 AM Mondays Office PGH 565 and Online 9:00 AM - 10:00 AM Fridays on Teams

General Course Information

(Optional: Enter Secondary Course Title)

Course Meeting Time: (Enter your class time)

Course Meeting Location: (Enter location or Delete this line)

Course Objectives and Student Learning Outcomes

Learning Objectives:

The objective is to prepare students such that within a few years of graduation they will be able to secure for themselves a successful computer science career and contribute decisively to the improvement and development of technology by demonstrating their ability to:

- Understand the context of neural networks and deep learning
- Know how to use a neural network
- Understand the data needs of deep learning
- Have a working knowledge of neural networks and deep learning.
- Explore the parameters for neural networks.

Learning Outcomes:

Based on satisfactory completion of the course, a student should be able to:

1. Identify the deep learning algorithms which are more appropriate for various types of learning tasks in various domains.
2. Implement deep learning algorithms and solve real-world problems.

THIS IS A REQUIRED COMPONENT (SECTION)
REQUIRED COMPONENTS MUST HAVE CONTENT ELSE THE SYLLABUS
CANNOT BE PUBLISHED.
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Required Instructional Materials

THIS IS A REQUIRED COMPONENT (SECTION)

EBook:

Neural Network Mastery By Nouhad Rizk

<https://he.kendallhunt.com/product/neural-network-mastery-journey-through-deep-learning>

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Course Schedule, Assignments, and Assessments

Detailed Course Schedule

Note: Schedule is tentative and may be adjusted as needed.

WeekTopicsLabsHomeworkProject / Assessments

Introduction to Deep Learning History and Applications	Lab 0: Crash Exercise (Ungraded)		Group Sign-Up
Logistic Regression Review Introduction to Deep Learning	Lab 1: Logistic Regression	HW1: Perceptron	
Neural Networks Fundamentals Perceptrons and MLPs	Lab 2: Neural Networks Lab 3: Backpropagation		Project Proposal Submission
Backpropagation and Training Neural Networks			
Convolutional Neural Networks (CNNs)	Lab 4: CIFAR-10		
CNNs and Introduction to RNNs	TensorFlow Introduction		Exam 1 (18%)
Recurrent Neural Networks (RNNs)	Lab 5: RNN	HW2: RNN	
LSTM Networks MNIST Applications Keras and TensorFlow			

ResNet Architectures Visualization and Adversarial Learning			
Advanced Deep Learning Topics GANs and Autoencoders		HW3: CNN	Project Presentation Discussion
Transformers			Exam 2 (18%)
Reinforcement Learning			Project Presentation Submission (PowerPoint/Video Demonstration)
In-Class Project Presentations			Poster Submission
Poster-to-Paper Conversion Workshop		HW4: RNN HW5: GAN	Paper Submission (Written Report)
Final Project Showcase			Presentation with the Data Science Institute



Discussion and Lecture Topics

Course Requirements:

1. Programming Assignments/ HW and Project (tentative) (14%+30%=44 %)

HW	Posted Wednesdays	Due Saturdays
HW 1	Mid of week	End of week 3 • 3%
Project Proposal	2-pages detailed abstract	End of week 4 • 7%

HW 2	Mid of week 5	End of week 7 • 3%
Project presentation	Ppt/video of the implementation	End of week 7 • 10%
	IN CLASS PRESENTATION	15 days before last day of class
HW 3	Mid of week 8	End of week 9 3%
HW 4	Mid of week 10	End of week 11 3%
HW 5	Mid of week 11	End of week 13 3%
Paper Submission	Written Report	Seven days before last day of class • 13%



2. Labs + Inclass (tentative) (10 %)

*Some of the python code outdated

Labs	Posted (Wednesdays)	Due (Saturdays)
Lab0	Mid of week 1	End of week 2
Lab1	Mid of week 2	End of week 3
Lab2	Mid of week 3	End of week 4

Lab3	Mid of week 5	End of week 6
Lab4	Mid of week 7	End of week 8
Lab5	Mid of week 9	End of week 11

3. CMAP (1.5 %)

CMAP	All Topics	Due (Tuesdays)
CMAP1	Topics from DS1	Mid of week 4
CMAP2	Neural network Forward/BACK propagation	Mid of week 6
CMAP3	CNN-RNN-GAN	Mid of week 9

4. CIA (1.5%)

CIA	All Topics	Due (Tuesdays)
CIA1	Topics from DS1	Mid of week 4
CIA2	Neural network Forward/BACK propagation	Mid of week 6
CIA3	CNN-RNN-GAN	Mid of week 9

Grading Rubrics and Weights

Grading Policy

The final numeric grade is computed based on student's performance in weekly assignments and exams/quizzes.

Plus and minus grading is used when determining final grades. Final grades are computed by first finding the average score in each category described in the table below on the right. All scores are normalized to a scale of 0 to 100 before being averaged. The average score for each category is then used to compute the weighted average according to the weights in the table.

Category % of Final Grade	
Attendance	1%
CIA +CMAP	3%
Lab work+ In class Quizzes	15 %
Homework Assignment (individual submission)	15%
Exams (September 29, November 3rd)	36 %
Project class Homework Assignments	30%
(Each grade= 50 % group grade + 50% *peer assessment)	
• Project Proposal (2 pages)	• 7%
• Project presentation (video)	• 10%
• Project Written report (paper)	• 13%

Grade % of Total Points

A $\geq$ 92.5 Excellent	A- $\geq$ 89.5 and $<$ 92.5 Outstanding	B+ $\geq$ 86.5 and $<$ 89.5 Very Good
B $\geq$ 83.5 and $<$ 86.5 Good	B- $\geq$ 79.5 and $<$ 83.5 Above Average	C+ $\geq$ 76.5 and $<$ 79.5 High Average
C $\geq$ 72.5 and $<$ 76.5 Average	C- $\geq$ 69.5 and $<$ 72.5 Low Average	D+ $\geq$ 65.5 and $<$ 69.5 Below Average
D $\geq$ 62.5 and $<$ 65.5 Poor	F $<$ 62.5 Failing	

Course Policies and Procedures

Important Notes

1. 3-Day Policy: One has 3 day starting from the end of the class time in which the graded

assignment/exam papers have been distributed and/or posted in order to object to the score

of that assignment or exam. The objection shall be submitted electronically by emailing the

TA and the instructor.

2. Academic Honor Code: As a student, you join a community of scholars who are committed

to excellence in learning. I assume that students will pursue their studies with integrity and

honesty. zero- tolerance for cheating, whether in exams, quizzes or programming assignments. Plagiarism, copying and other anti- intellectual behavior are prohibited by the university regulations. Violators will face serious consequences.

3. Student Conduct: Disruptive behavior inside or outside class may result in disciplinary actions and academic failure. Students must refrain from disturbing the peace and good order of the university. For more details, please refer to

<http://www.uh.edu/dos/pdf/codeofconduct.pdf>

4. Academic Integrity: Cheating or any other suspected violations of academic integrity will

not be tolerated and will be reported to the Department of Computer Science, Director of Undergraduate/Graduate Studies and if substantiated may result in significant penalty. It is

each student's responsibility to read and understand the Academic Honesty Policy found in

the Student Handbook (<http://www.uh.edu/academics/catalog/policies/academic-reg/academic-honesty/>).

5. Plagiarism: Plagiarism is using someone else's work without proper acknowledgement.

This includes getting help from a friend or colleague and online material. When using someone else's work, always cite the source. Plagiarism is considered a serious breach of academic integrity. Any breach of academic integrity or plagiarism would result in a minimum of one full letter grade reduction over the final score and possible expulsion from

university.

6. Attendance: Attendance is strongly recommended. Students with unexcused absences should not expect additional help outside of class and are still responsible for any material or instructions given in class, for turning in assignments on time, and for taking exams at the scheduled times.

7. Make Up: NO Make up policy. Missed exams might only be considered (the excused absence does not guarantee exam scheduling) for excused absences with advanced notice (BEFORE THE EXAM) of at least three working days with full documentation. A grade of

zero will be assigned for missed exams, if documentation has been submitted within one day

or the day of the exam or after the exam date.

8. Reading: Readings from the textbook will be given out throughout the semester. We will not

be able to discuss all of it in class, but you are still responsible for keeping up with these readings.

9. Code Documentation: When assignments are graded, source code may be examined to verify the way a solution was achieved or to award partial credit. It is your responsibility to

make sure that your source code is presented in a clear, readable, way.

10. NO LATE POLICY ON ANYTHING=> nothing will be accepted late!

University Policies and Student Support Resources

Mental Health and Wellness Resources

The University of Houston has a number of resources to support students' mental health and overall wellness, including [CoogsCARE](#) and the [UH Go App](#). [UH Counseling and Psychological Services \(CAPS\)](#) offers 24/7 mental health support for all students, addressing various concerns like stress, college adjustment and sadness. CAPS provides individual and couples counseling, group therapy, workshops and connections to other support services on and off- campus. For assistance visit uh.edu/caps, call 713-743-5454, or visit a [Let's Talk](#) location in-person or virtually. Let's Talk are daily, informal confidential consultations with CAPS therapists where no appointment or paperwork is needed.

Need Support Now? If you or someone you know is struggling or in crisis, help is available. Call CAPS crisis support 24/7 at 713-743-5454, or the National Suicide and Crisis Lifeline: call or text 988, or chat 988lifeline.org.

Title IX/Sexual Misconduct

Per the UHS Sexual Misconduct Policy, your instructor is a "responsible employee" for reporting purposes under Title IX regulations and state law and must report incidents of sexual misconduct (sexual harassment, non-consensual sexual contact, sexual assault, sexual exploitation, sexual intimidation, intimate partner violence, or stalking) about which they become aware to the Title IX office (known at UH as the Equal Opportunity Services office or "EOS"). Please know there are places on campus where you can make a report in confidence. You can find more information about resources on the UH [Title IX/Sexual Misconduct Resources page](#). Please note that you may also report concerns of discrimination based on your protected class identity to EOS.

Reasonable Academic Adjustments/Auxiliary Aids

The University of Houston is committed to providing an academic environment and educational programs that are accessible for its students. Any student with a disability who is experiencing barriers to learning, assessment or participation is encouraged to contact the Justin Dart, Jr. Student Accessibility Center (Dart Center) to learn more about

academic accommodations and support that may be available to them. Students seeking academic accommodations will need to register with the Dart Center as soon as possible to ensure timely implementation of approved accommodations. Please contact the Dart Center by visiting the website: <https://uh.edu/accessibility/> calling (713) 743-5400, or emailing jdcenter@Central.UH.EDU.

The [Student Health Center](#) offers a Psychiatry Clinic for enrolled UH students. Call 713-743-5149 during clinic hours, Monday through Friday 8 a.m. - 4:30 p.m. to schedule an appointment.

The [A.D. Bruce Religion Center](#) offers spiritual support and a variety of programs centered on well-being.

The [Center for Student Advocacy and Community \(CSAC\)](#) is where you can go if you need help but don't know where to start. CSAC is a "home away from home" and serves as a [resource hub](#) to help you get the resources needed to support academic and personal success. Through our [Cougar Cupboard](#), all students can get up to 30 lbs of FREE groceries a week. Additionally, we provide 1:1 appointments to get you connected to on- and off-campus resources related to essential needs, safety and advocacy, and more. The [Cougar Closet](#) is a registered student organization advised by our office and offers free clothes to students so that all Coogs can feel good in their fit. We also host a series of cultural and community-based events that foster social connection and helps the cougar community come closer together. Visit the CSAC homepage or follow us on Instagram: @uh_CSAC and @uhcupbrd. YOU belong here.

Cougar Parent and Family Engagement Office

This office, under the Dean of Students Office (<https://www.uh.edu/dos/>), coordinates resources for Pregnant and Parenting Students (<https://www.uh.edu/dos/parenting-students/>), Parents and Family members of UH students (<https://www.uh.edu/dos/parents/>), and Sexual Misconduct Support Services (<https://www.uh.edu/dos/sexual-misconduct-support-services/>). A description of available support and resources are provided on the respective web pages. The Cougar Parent and Family Engagement Office is located in Student Center South, room B12.

Academic Honesty Policy

High ethical standards are critical to the integrity of any institution, and bear directly on the ultimate value of conferred degrees. All UH community members are expected to contribute to an atmosphere of the highest possible ethical standards. Maintaining such an atmosphere requires that any instances of academic dishonesty be recognized and addressed. The [UH Academic Honesty Policy](#) is designed to handle those instances with fairness to all parties involved: the students, the instructors, and the University itself. All students and faculty of the University of Houston are responsible for being familiar with this policy.

Excused Absence Policy

Regular class attendance, participation, and engagement in coursework are important contributors to student success. Absences may be excused as provided in the University of Houston [Undergraduate Excused Absence Policy](#) and [Graduate Excused Absence Policy](#) for reasons including medical illness of student or close relative, death of a close family member, legal or government proceeding that a student is obligated to attend, recognized professional and educational activities where the student is presenting, and University-sponsored activity or athletic competition. Under these policies, students with excused absences will be provided with an opportunity to make up any quiz, exam or other work that contributes to the course grade or a satisfactory alternative. Please read the full policy for details regarding reasons for excused absences, the approval process, and extended absences. Additional policies address absences related to [military service](#), [religious holy days](#), [pregnancy and related conditions](#), and [disability](#).

Recording of Class

Students may not record all or part of class, livestream all or part of class, or make/distribute screen captures, without advanced written consent of the instructor. If you have or think you may have a disability such that you need to record class-related activities, please contact the [Justin Dart, Jr. Student Accessibility Center](#). If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Classes may be recorded by the instructor. Students may use instructor's recordings for their own studying and notetaking. Instructor's recordings are not authorized to be shared with anyone without the prior written approval of the instructor.

Failure to comply with requirements regarding recordings will result in a disciplinary referral to the Dean of Students Office and may result in disciplinary action.

GenAI use

Acceptable Use for Students

Students may use Generative AI only in ways that support learning, not replace it. Students are expected to:

Use AI as a tool for brainstorming, feedback, practice, and clarification

Follow assignment-specific instructions on when and how AI is allowed

Verify facts, sources, and citations provided by AI

Acknowledge and disclose AI use when required by the instructor. Students are not permitted to:

Submit AI-generated work as their own when prohibited

Use AI during assessments unless explicitly allowed

Rely on AI to complete work they are expected to do independently

Use AI tools that violate school privacy or data-use policies

Misuse of AI will be addressed under existing academic integrity and conduct policies.

Phone or any electronic device

Phones or any electronic device are not permitted to be used at any time during class or during exams. During exams, students are required to turn off their phones and place them face down on their desks for the entire duration of the exam. Any use of a phone during an exam—including checking, handling, or accessing the phone for any reason—will

be considered cheating and may result in disciplinary action in accordance with university policy.

Late Policy

No-Late-Submission Policy: Any assignment submitted after the stated deadline will receive a grade of zero.

No-Exam-Makeup Policy: Any missed examination will receive a grade of zero. Makeup examinations will not be offered.

Important Dates

University Holiday

- Labor Day: Monday, September 1

Exams

- Exam 1: 18% September 29
- Exam 2: 18% November 3rd

Project Deliverables first week of december (check with DSI)

DeliverableExpected Week

Group Sign-Up	Week 1
Project Proposal	Week 3
Project Presentation Discussion	Week 7
Project Presentation Submission (PowerPoint/Video)	Week 7
Poster Submission	Week 13
Paper Submission (Written Report)	Week 14
Final Presentation with Data Science Institute	Week 15

Course Components

Homeworks

- HW1: Perceptron
- HW2: RNN
- HW3: CNN
- HW4: RNN (Extended Analysis)
- HW5: GAN

Labs

- Lab 0: Crash Exercise (Ungraded)
- Lab 1: Logistic Regression
- Lab 2: Neural Networks
- Lab 3: Backpropagation
- Lab 4: CIFAR-10
- Lab 5: RNN

Deep Learning Topics Covered

1. Logistic Regression Review
2. Neural Networks and Backpropagation
3. CNNs
4. RNNs and LSTMs
5. Keras and TensorFlow
6. ResNet
7. Adversarial Learning
8. GANs
9. Autoencoders
10. Transformers
11. Reinforcement Learning

This version presents a much clearer progression from foundations → neural networks → CNNs → RNNs → advanced deep learning → project deployment and presentation.

Class Project Guidelines

The class project is a major component of your final grade. It is designed to give you hands-on experience applying course concepts to a real-world, creative problem that you can showcase on your resume or portfolio.

1. Group Structure & Grading Rules

- Group Size: Teams consist of 3 students (formed by you, or randomly assigned by the course staff if needed).
- Evaluation by Size: Project scope expectations scale with group size—a project suitable for 3 students will not meet the bar for a larger group.

- Shared Grade: All group members receive the same grade. Choose your teammates carefully, establish regular communication (via Slack, Email, etc.), and divide computational and writing tasks evenly.
- Grading Focus: Grades are based on your learning process, execution, and honest reporting—not on whether your model achieves high performance. If an experiment fails or yields low accuracy, you will not be penalized, provided you analyze the issues clearly and suggest future improvements.

2. Project Deliverables

Your project consists of three main parts:

Part 1: Project Proposal

- Purpose: Get early feedback from instructors/TAs on project feasibility and scope.
- Format: 1–3 pages (800–1,200 words, excluding references). Must be written using the provided template and submitted as a PDF. Figures are encouraged.
- Required Sections:
 1. Introduction: What you plan to do and relevant background/related work.
 2. Motivation: Why this project matters (societal impact, practical utility, or personal learning goals).
 3. Evaluation: Define what a "successful" outcome looks like and how you will measure it technically.
 4. Resources: Datasets, hardware, and computational tools you plan to use.
 5. Team Contributions: Clear breakdown of each member's specific coding, experimental, and writing responsibilities.
- Note: You may update your project approach later in the semester if you learn new methods in class, but check with a TA first.

Part 2: Class Presentation

- Recorded video of the powerpoint of your project implementation. **This is an expansion of all poster template sections.** Format: 8–10 minute presentation + 2 minutes for Q&A.
- Timing: Held during the last five class lectures. Order of presentation is chosen at random each day.
- Participation: All group members must present a section.
- Required Content:

- Introduce the topic to a general audience.
 - Summarize your method/approach.
 - Highlight key outcomes and findings.
- Tech Setup: Bring your own laptop. Display adapters (HDMI, USB-C, DisplayPort, Lightning) will be provided.

Part 3: Final Project Report

- Format: 6–8 pages (excluding references).
- Reproducibility Standard: The report must be detailed enough that a classmate could replicate your results without guesswork.
- Required Sections:
 1. Introduction
 2. Related Work
 3. Proposed Method
 4. Experiments
 5. Results and Discussion
 6. Conclusions
 7. Team Contributions (Detailed breakdown of individual work)

Part 4: Poster template

			[Insert Title] [Insert Authors]	
[Insert Department], University of Houston				
Abstract Insert abstract here		Methods Insert methods here		Results Insert results here
Background Insert background here		Insert figure here		Conclusion Insert conclusion here
				Future Direction Insert future direction here
				Acknowledgments Insert acknowledgements here

You can view examples of previous submissions here: [Scholar Excellence in Research](#).

Additionally, please attend the [AI in Medicine Seminar](#).

This is a great opportunity for you to network and potentially collaborate on your projects with industry professionals and researchers.

The project is mandatory and you are required to present at the HP Data Science Institute at the end of the semester. (More here : <https://hpedsi.uh.edu/partnerships/sponsored-events/data-science-showcase>)